Technology in Phonetic Science: Setting Up a Basic Phonetics Laboratory

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Abstract

The present paper discusses some techniques and procedures used in experimental phonetic research. The use of technology and its recent widespread availability have been instrumental in our developing a better understanding of the basic processes of human speech production and perception. Specifically, this paper reviews some of the uses of the software package Praat for acoustic and perceptual research, as well as software products such as R, Akustyk, SuperLab, and E-Prime, among others. Finally, we offer some remarks on the basic equipment (hardware) one would need to purchase in order to carry out modern experimental phonetic research in the field as well as in the laboratory.

1. Introduction

There is a long-standing distinction between the fields of *impressionistic* and *experimental* phonetics. Hayward (2000) explains this as follows:

> Traditionally, phoneticians have relied on their ears and eyes, and their awareness of their own vocal organs, to study pronunciation. Increasingly, however, they have been using instruments of various types to supplement the information they derive from their own sensation. *Experimental phonetics*, as the term is commonly used, includes any investigation of speech by means of instruments. It is understood here that the instruments are used to visualize some aspect of the speech event, and possibly also to provide a basis for measurements (2000, p.1).

Decades of research on speech perception may serve to warn us that "sensations" are not fully reliable, including those of phonetically trained individuals, hence the need for instruments that allow researchers to "visualize" speech in an objective (rather than subjective), quantifiable and replicable way. The use of technology by itself, however, does not make phonetic research experimental. The term "experimental phonetics" implies that experiments are used. Experiments are contrived observations. First, the experimenter designs a deliberate plan that will trigger capturable observations (data). Second, he or she reveals the details of the design to the scientific community so that others may replicate it at their own discretion. Third, the experimenter acquires and stores the observations in such a way that they are quantifiable and may be submitted to statistical exploration, which is the standard in modern scientific research.

Experimental linguistic research has been growing at an exponential rate in recent decades. While syntax, semantics (cf. Gibson & Fedorenko forthcoming) and phonology (cf. Pierrehumbert, Beckman & Ladd 2000) scholars have seen an increase in experimental methods, these have been the standard, dominant practice in phonetic (speech production and perception) as well as psycholinguistic (i.e. language processing) research for some time. My personal viewpoint is that there is no way back. Therefore, students considering a career as linguists need to seriously consider jumping on the bandwagon of instrumentation, experimentation and quantification.

The present article offers a few remarks on the types of instruments that are being used in basic phonetic research. Of course, one cannot possibly review all the existing instruments. This article's selection, thus, is biased and skewed to include what I personally have used or know best. My selection will leave out many instruments others might consider fundamental as well as some I would like to include. This is inevitable. Sorry. Finally, let me add that I decided to focus on instruments that I consider relatively unsophisticated and generally affordable. Others require a great deal of specialized knowledge, are very expensive and/or are available in just a handful of laboratories. What I discuss here is available to most practitioners with a modest budget.

2. Speech production

The result of the motor activity of the articulators is an acoustic signal or sound wave. The acoustic signal may be studied with fairly standard and robust methods developed in the fields of physics and electrical engineering. Although it is known that the relationship between the sound wave and the dynamics of the articulators in all their spatial detail is not fully linear, it is also clear that there is a non-random relationship between the two (e.g. Stevens 1972, 1989, 1998). For instance, Iskarous (2010) shows that "distinctive articulatory information for vowels can be recovered from the acoustic signal" (Iskarous 2010, p.375). This is a limitation of acoustic phonetics that scholars need to bear in mind at all times. However, studying the acoustic signal *in lieu* of the movements of the articulators has a number of advantages. It is easy and relatively cheap to obtain a good quality signal from which to derive quantitative measurements. Acoustic signals may be obtained from a large pool of speakers, in the field or in the laboratory and does not require invasive methods. Therefore, although the study of the acoustic signal has some serious limitations if one wishes to derive inferences regarding the speech production system, the truth is that it has been widely used for that purpose and it is unlikely that this is going to change soon. Phoneticians wishing to pursue this method should be aware of its limitations and should be willing to investigate the existence of non-linearities between acoustics and articulation as parallel data become available or to at least interpret their acoustic findings with care.

This section is not an introduction to acoustic phonetics or to analysis methods for speech sounds (see, among others, Harrington 2010, Hayward 2000, Johnson 2003, Ladefoged 2003, 2004, Ladefoged & Johnson 2006, Reetz & Jongman 2008). I will provide a brief and inevitably biased overview of some recent examples. Additionally, I will also exemplify how these measurements could be made with Praat (Boersma 2001). Praat is a powerful, free and multi-platform digital signal processing software package (www.fon.hum.uva.nl/praat/). Praat generates oscillograms, spectra and spectrograms of digital sound files. These may be viewed in Praat or may be exported to external EPS or PDF files, which can then be pasted on articles or theses. Praat performs formant, pitch and intensity analyses. It may be used for speech synthesis and manipulation. It has an enormous array of built-in functions, which allow the researcher to obtain a measurement by simply clicking on a button. Finally, it is scriptable. This is very important for the experimental phonetician. Suppose that you need to obtain formant values (e.g. F2) from 1000 vowel tokens. Would you be willing to click on the same button 1000 times and then write down 1000 numbers? In Praat, one may write a brief script that will obtain the 1000 measurements automatically and save them on a tab-delimited table file. This text file may then be exported onto a spreadsheet program such as Excel or LibreOffice or directly to a statistics or mathematics software package, such as SPSS, R, Matlab or Octave. Let us see some examples.

Escudero, Boersma, Rauber & Bion (2009) carried out an acoustic analysis of Portuguese vowels recorded by 40 speakers. An impressive total of 5600 tokens were collected. Their goal was to compare the vowels of two dialects (Saõ Paulo and Lisbon) and to investigate how constraints that seem to affect vowel systems in general may affect the Portuguese vowel system in particular. Escudero et al. performed a static analysis of three spectral parameters (F0, F1, and F2) as well as an investigation of the duration patterns of the vowels. Theirs was a static analysis in the sense that spectral data were extracted from a single point in each of the vowels. In particular, spectra were generated from the central 40% of the vowel. This means that the window was longer for longer vowels and shorter for shorter vowels. Although it is standard to derive spectral information from vowel midpoints, it is not standard to extract them from variable-length windows. Praat's standard for formant analysis appears to be a fixed 25 ms analysis window, but that may be easily changed in Praat and different researchers use different window lengths.

F0 values may be obtained from Praat's Pitch object. (I recommend the interested reader to search for information on the Praat objects I mention in Praat's searchable Help manual). At least two methods are available in Praat, the autocorrelation and the cross-correlation algorithms. The Pitch object, once generated, has a number of options. For instance, one may get the pitch value of a specific point in time in the sound file. Pitch values may be calculated in Hz, logHz, Mel, ERB and Semitones. Praat will automatically choose the nearest pitch value or carry out a linear interpolation between the two nearest points if one specifies this option. Escudero et al. decided to extract the median pitch value of the central 40% of the vowel; thus, they extracted all pitch values in that region 1 ms steps and then extracted the median. According to the authors, this is more robust than calculating the mean pitch of the vowel or than extracting a single, fixed pitch value from the vowel's midpoint.

Formant values may be obtained from Praat's Formant object. Praat's standard formant tracking algorithm is the Burg algorithm. Other algorithms are available. A Formant object offers many options, one of which is to provide the desired formant value (F1 to F5) of a particular time point in either Hz or Bark. Anyone who has ever used the Praat-standard built-in function (extract 5 formants in the range 50 Hz to 5500 Hz for females and 50 Hz to 5000 Hz for males) will have been able to observe numerous tracking errors on the screen (see discussion in Escudero et al. 2009). If one changes the parameters, one gets a slightly different result. If we are to rely on automatic measurements, we should try to minimize the likelihood of these tracking mistakes. There are a number of methods to reduce noise, but none of them will altogether eliminate it. Doing everything manually is likely to introduce an even greater amount of noise. Ultimately, this is the researcher's choice. Escudero et al. (2009) devised a method in which a formant object for a given vowel token would be generated with its "optimal ceiling." Optimal ceilings were calculated per vowel category per speaker (there were 20 vowel tokens per category per speaker in their data set). The authors wrote a Praat script that generated 201 formant objects per each vowel token. Each one of the formant objects had a different ceiling parameter; that is, the procedure extracted two formants from formant objects with varying ceiling parameters in the range 4500-6500 Hz (for females) or 4000-6000 Hz (for males) in steps of 10 Hz. An "optimal ceiling" was the one that produced the lowest variance for the distribution of the 20 data points calculated with this ceiling parameter. This resulted in the selection of 280 "optimal ceilings," one per vowel category per individual speaker. In all, Escudero et al. calculated 5600 * 201 = 1125600 formant objects and extracted 11125600 * 2 (formants) = 22501200 formant values. Needless to say, this procedure would have simply been impossible if Praat were not a scriptable program. With this example one can easily understand why having access to powerful, scriptable digital speech processing software is necessary in modern phonetic research.

When a researcher wishes to compare vowels across speakers, he or she finds the following problem: formant values are affected by numerous non-linguistic constraints, such as vocal-tract size. Therefore, a given research finding (statistical difference) might be due to a number of reasons, many of which will be non-linguistic and thus irrelevant to many researchers. We will refer to this as the "normalization problem." In order to reduce the effect of these "irrelevant" factors, a number of linguists have devised different normalization procedures. A normalization procedure warps the vowel space of a speaker so that it makes it resemble that of another speaker. In this way, between-speaker comparisons may be carried out in a manner that eliminates or reduces anatomical/physiological effects but retains the "desired" differences, such as structural, sociolectal or geolectal differences.

Adank, Smits & Van Hout (2004) and Fabricius, Watt & Johnson (2009) recently evaluated several normalization procedures. According to these authors, the most robust procedures for sociophonetic research (i.e. the ones that retain structural and dialectal information while eliminating anatomical/physiological effects) are vowel and formant extrinsic and speaker intrinsic. Consider, for instance, the Lobanov z-score procedure (Lobanov 1971). In this procedure, a particular formant value is submitted to the following formula: *F (norm) = (F (Hz) – M (Hz)) / SD (Hz)*. In other words, the procedure first calculates the Hz distance between a particular formant value (*F (Hz)*) and the speaker's vowel space centroid for the same formant (*M (Hz)*) (i.e. across all the vowels in the vowel space) and then divides that number by the standard deviation of the speaker's vowel space (*SD (Hz)*). This is a speaker-intrinsic procedure because all the values that go in the equation come from the same speaker. It is vowel extrinsic because a particular vowel token is compared to averages and standard deviations extracted from the entire vowel system (i.e. all the gathered vowel tokens of all categories). Finally, it is formant intrinsic because F1 values enter the formula exclusively with other F1 values and F2 values enter the formula exclusively with other F2 values. This procedure is useful only when it may be assumed that all subjects have the same vowel system; that is, only to compare a pool of speakers of the same language or across languages that are supposed to share a vowel inventory.

An example of a very different procedure is provided in Guion (2003). Guion extracted formant values in Bark, and not Hz. These were used for all within-speaker comparisons. For between-speaker comparisons, on the other hand, Guion normalized the Bark values using the following formula: *F (norm) = F (Bark) Si * (14.297 / M F3 (Bark) Si)*. The formant values in Bark (both F1 and F2) of each vowel token of speaker *Si* were multiplied by a factor obtained by dividing 14.297 (the average F3 of one male speaker to whom all other speakers were normalized) by the mean F3 of speaker *Si*. This procedure is formant extrinsic because it compares F1 and F2 values to F3 data. It is speaker extrinsic because it introduces in the equation data from different speakers. F3 enters the equation because apparently it provides a good measure of the size of the vocal tract of the subject. This transformation has the advantage of being appropriate when comparisons across speakers with different vowel systems are needed, which was the case in Guion (2003).

A third way of dealing with the "normalization problem" is to avoid it altogether. This was the decision made by Tsukada, Birdsong, Bialystok, Mack, Sung & Flege (2005). Among other things, Tsukada et al. investigated the production differences of an English-specific vowel contrast by several groups of Korean-English bilinguals (including adults and children) and native English speakers. The authors reasoned that anatomical/physiological differences across the participants could be too large in this case and thus decided to not compare vowel categories across speakers. Instead, they decided to compare the degree to which a phonetic contrast was maintained by each of the subjects. They calculated the mean F0, F1 and F2 values in Bark of the vowels involved (these were extracted from LPCs generated from 20 ms, fixed-length windows centered at the vowel's midpoint). The within-vowel F1-F0 and F2-F1 distances were calculated in order to obtain a slightly transformed measure of vowel height and fronting. The authors then calculated the Euclidean distance (on the height * fronting space) between the two vowel categories of a particular speaker. A large Euclidean distance signaled that a robust contrast was maintained between the two vowels for that speaker while a small Euclidean distance showed that a phonetic contrast was reduced or even not at all present. This procedure does not completely avoid the "normalization problem" but it drastically reduces it because it does not compare vowel categories across subjects but rather within-speaker, between-vowel distances.

I would say that, by now my point is quite clear. Can you imagine calculating all these values by hand? Would you be willing to submit all your F1 and F2 values to complex formulas if you knew you had to calculate them with a desktop calculator? The use of software packages that allow for these calculations in a matter of seconds is a must in modern phonetic science. The mathematical transformations entailed in between-speaker normalizations, although not completely problem-free, significantly reduce the chance of reaching erroneous inferences and should be seriously considered by all experimental phoneticians wishing to compare data from different speakers. For instance, imagine that one has extracted F1 and F2 values from 1000 vowel tokens and has them saved on a tab-delimited text table. Each row contains the information of a particular vowel token in 7 columns: token number, speaker, gender, dialect, vowel phoneme, F1 value and F2 value, both in Hz. Praat would be able to read such a text table as a Table object. One should then write a script that would read the formant values of each one of the vowel tokens (columns 6 and 7), perform the selected transformations (one can carry out mathematical functions with Praat scripts) and paste the resulting, normalized values to the table, e.g. in columns 8 (for normalized F1) and 9 (for normalized F2). Praat also has built-in functions that carry out some common transformations without one having to write the formula (search for "mathematical functions" in the Help manual): *hertzToBark*, *hertzToMel*, *hertzToErb*, etc.

The need for fast, automatic procedures is even clearer when one wishes to investigate the dynamics of speech production. Dynamic analyses, unlike static ones, take into account the nature of the spectral differences between two or more time frames. Speech is a dynamic phenomenon and even very small changes in the shape of the spectral envelope may be linguistically relevant. For instance, consider the difference between a monophthong and a diphtong. A diphthong differs from a monophthong not only (or not mainly) on the absolute formant values extracted from the midpoint of the vowel or vowel sequence but on the direction and degree of change of the formants as the vowel unfolds in time. Measurement procedures have been developed to capture the dynamics of formant values in vowels and vowel sequences. This is an area, however, that merits even more attention than it has received in the past. I expect that scholars will continue to come up with even more powerful ways to capture change in time in their data (e.g. Harrington, Kleber & Reubold 2008).

Aguilar (1999) provides an example. Aguilar researched the acoustic differences between phonological hiatuses and diphthongs as well as the effects of speech style on both categories. In addition to differences in duration, Aguilar hypothesized that the dynamics of the formants would differ between the two syllabification configurations. In order to capture formant change, Aguilar calculated a series of LPC spectra of 20 ms fixed-length windows in 10 ms left-to-right steps. In other words, F1 and F2 were calculated every 10 ms from the beginning to the end of the vowel sequence. This allowed for the tracking of the formants as vowels unfold. Subsequently, formants tracked in this way were fitted one by one by quadratic or second-order polynomial equations. Second-order polynomials have a curvature coefficient, a slope coefficient and a y-intercept. Aguilar then submitted the coefficients to statistical analyses. The results showed that hiatuses have a greater degree of curvature than diphthongs; that is, diphthongs have a more smooth change in their formants configuration than hiatuses. It is obvious that extracting F1 and F2 at a fixed time point in the vowel sequence would not have been able to capture the dynamic nature of the difference between these two categories.

Andruski & Costello (2004) and Simonet (2009, forthcoming) applied this curve-fitting procedure to the study of pitch curves. Andruski & Costello explain how one might carry out such an analysis with Excel. The following is a simplification of the procedure used in Simonet (2009, forthcoming), where Praat and R were used. R is a powerful, free and multi-platform language and environment for statistical exploration and graphics (www.r-project.org). A large community of users writes functions that are made available to the public in the form of packages that one may download and install for free. First, one might delimit the time span (area) from which formant (or pitch) points will be tracked. This may be done with Praat's TextGrid object (see below). This will have to be done for as many vowel tokens one wishes to measure. Second, one writes a script that proceeds on a left-to-right step-wise manner and tracks the spectral values (e.g. F1 or F2). For instance, one may write a procedure that will generate 20 ms fixed-length LPC spectra every 10 ms. This information is stored in a separate text file (Praat's Table object) with two columns: time, formant value. If these two columns are plotted on a bi-dimensional plot, one obtains a reproduction of the formant track. If we have a data set comprised of a total of 1000 vowels, we end with a list of 1000 two-column text files (or Table objects). The third step is to write an R script that will read the two-column text files one by one in R, calculate the best fit quadratic polynomial equation of the data points and generate a new multi-column text file (or data frame in R's jargon). The date frame has as many rows as vowels there are in the data set. Each row contains the information of a single vowel, with each bit of information in a separate column: token number, speaker, gender, dialect, phonological category, a-coefficient (curvature), b-coefficient (slope) and c-coefficient (y-intercept). This data frame, when completed, may then be read in R for statistical exploration.

Let us now proceed to a discussion of some acoustic measurements used in the analysis of stop consonants. A standard acoustic parameter that has been used to capture the nature of the difference between utterance-initial voiced and voiceless stops is the time latency between the onset of phonation or glottal pulsing and the time of articulatory release. This is called VOT, for voice onset time (e.g. Lisker & Abramson 1964). This is a very straightforward acoustic parameter – it may be measured easily. Notice Flege & Eefting's (1988) explanation of how they carried out such measurements not that long ago: "voice onset time was measured using a ruler to the nearest 0.5 mm (2.0 ms) from the beginning of the release burst to the onset of periodicity in the region of the second formant in the following vowel (in voiceless unaspirated and aspirated stops), or from the beginning of low-frequency striations to the beginning of the release burst (in prevoiced stops)" (Flege & Eefting 1988, p.732). Let me say that you will no longer need to measure VOT using a ruler to the nearest 0.5 mm. How did they manage to measure 5600 vowel tokens from 70 speakers using this procedure?

What can you do with Praat nowadays? One has at least two options. First, one may manually identify the beginning of the two temporal events pertaining to the VOT measure with the help of Praat's time-aligned text files (TextGrid object). TextGrid objects are text files that align with sound files. They may be organized in different tiers, with each tier being dedicated to a different event. One can introduce annotations in the tiers and can mark time events. In order to measure VOT one would need to open the sound files one by one, generate a two-tier time-aligned TextGrid, mark the release of the articulators on one tier by placing the cursor on the event and then clicking on the TextGrid to annotate it, and then mark the onset of phonation in the same way. Since the two events will be introduced on two different tiers, one allows for one event preceding the other in some consonants and following it in others, just as it happens with /p/ or /b/ in different languages. At the end of this manual introduction of data one can write a script that will extract the time values of the two events in each of the consonant tokens and calculate the time difference between the two. These measurements could then be saved on a multi-column Table object with the following structure: token number, speaker, speaker group, phonological category and VOT (ms). Thus, a procedure that used to take weeks or months may now be reduced to a single day or even a few hours.

But we could go even further in our automatization endeavors. Fowler, Sramko, Ostry, Rowland & Halle (2008) use the following automatic procedure: an assistant placed a boundary to the left of the release of the articulators and another one to the right of the onset of glottal pulsing. This does not seem to differ much from what was explained in the preceding paragraph, but it does. Notice that the previous procedure required that one manually identified the two acoustic events. The difference in ms between those two manual marks *is* the VOT measurement. In Fowler et al.'s procedure, one only introduces two boundaries that will later be used to identify the two events automatically. The two manual marks are not themselves used in the calculations. This significantly reduces human error. Here is how it works. After the two boundaries have been identified, a script identifies "RMS amplitude on a per sample basis using a moving rectangular window 4 ms in length. Burst onset was the first sample that preceded voice onset by 8 ms or more and had a magnitude that exceeded 40% of the RMS maximum. Voice onset was defined as the first sample that exceeded 50% of the maximum RMS for that token" (Fowler et al. 2008, p.655). A very similar procedure could be carried out entirely with Praat's IntensityTier (dB SPL) or AmplitudeTier (Pascal) objects and their built in functions. The procedure would not differ much from the tracking of formant or pitch trajectories explained above as one would need to write a script that extracts amplitude values on a per sample basis and stores that information. The difference is that here the script is looking for a specific time point and, when it finds it, it stores that information only and forgets the rest.

Let us now consider a much more slippery acoustic parameter: articulatory lenition. At present, very limited articulatory-acoustic parallel data has been gathered to verify whether the acoustic metrics that have been used so far are reliable (cf. Parrell 2010). Most of these methods rely on intensity estimates of the signal. Praat's Intensity and IntensityTier objects may be used in this regard. For instance, one could measure the difference between the intensity minimum (i.e. maximum constriction degree during the consonant) and the surrounding intensity maxima (i.e. maximum opening degree of the pre- and post-consonantal vowels). Perhaps, one could consider the ceiling to be the mean intensity of the two surrounding maxima. Another option is to calculate the ratio between these two intensity values. A third option requires some more calculations. Hualde, Simonet & Nadeu (forthcoming) and Kingston (2008) calculated the first-difference curve of intensity tracks. The curve was calculated in 1 ms steps so that they obtained a representation of the velocity contour of the intensity track. Subsequently, an automatic Praat procedure found the velocity peak between the intensity minima and the following intensity maxima. The height of the velocity peak was interpreted to be a correlate of constriction or lenition degree, i.e. a low peak occurs in smooth transitions (and thus lenited consonants) while a high peak occurs in abrupt transitions (and thus very constricted consonants). Hualde, Simonet & Nadeu (forthcoming) found that the three procedures were highly correlated with each other. I simply can't envision a way in which such a procedure would be carried out manually, i.e. without access to computing equipment.

This section has reviewed some basic procedures to carry out acoustic measurements in vowels and stops consonants (as well as in pitch contours). My main goal has been to show that these methods have been made easy and accessible in recent years through the advent of powerful and free computing tools, such as Praat and R. It would seem wise to invest some time to learn these tools, which shorten the time dedicated to data analysis and improve the quality and reliability of the measurements. In many cases, these tools make possible analyses and metrics that would not be possible without them.

3. Speech perception

The present section briefly discusses the role of technology in perception and sound processing experiments. In particular, we review two recent papers that present two basic methods. There are numerous software packages designed specifically for behavioral psychological experiments. They can all be used to carry out perception research. These programs include E-Prime, SuperLab, Presentation, Paradigm, PsyScope X, DMDX and PsyToolkit. Some of these programs are commercial and some are free. Some work exclusively on Windows, exclusively on Mac OS X or exclusively on Linux. SuperLab runs on both Mac OS X and Windows. Many other tools are available. For instance, Praat may run most types of perception experiments with its ExperimentMFC object. Let us now see two examples of what one might do with these software tools.

Pallier, Bosch & Sebastián-Gallés (1997) investigated the perception of a Catalan-specific vowel contrast by Spanish-Catalan bilinguals. In particular, they examined the behavior of two groups of bilinguals, Catalan-dominant listeners and Spanish-dominant listeners. Their procedure was based on the categorical perception paradigm (Liberman, Harris, Hoffman & Griffith 1957). Liberman et al. generated an artificially-manipulated /ba/-/pa/ continuum and showed that English listeners categorized the lower half of the continuum mostly as /ba/ and the higher half of the continuum mostly as /pa/. This showed that perception of speech is affected by phonological categories. Pallier, Bosch & Sebastián-Gallés (1997) hypothesized that Catalan-dominant listeners would display categorical perception (effects of phonological categories in their perceptual behavior) while Spanish-dominant ones would not. Pallier, Bosch & Sebastián-Gallés generated an artificial seven-step continuum between two Catalan vowels and asked listeners to categorize or identify the stimuli as belonging to either one or the other vowel phoneme. Their results showed that Catalan-dominant listeners classified the lower part of the continuum mostly as one category and the higher half mostly as the other category.

Spanish-dominant listeners displayed a flat behavioral contour; that is, their classification of all stimuli as one or the other vowel phoneme was systematically at chance. The next step was to examine the perceptual discriminatory abilities of the bilingual subjects. The participants were asked to listen to pairs of stimuli that would either be identical or would differ by two steps on the continuum (s1-s3, s2-s4, etc.). Their task was to respond whether they thought the two sounds were the same or different. It has been found that listeners are reliably capable of hearing acoustic differences between sounds that are both located in the middle of the continuum while, when both sounds occur towards one of the two extremes, listeners tend to not hear the difference between the two and thus respond 'same.' Pallier, Bosch & Sebastián-Gallés indeed found that Catalan-dominant listeners displayed a peak in their discrimination rates towards the middle of the continuum while Spanish-dominant listeners displayed, once again, a flat contour. This was evidence of the effect of categorical perception in one group of bilinguals but not the other.

The first issue that comes to mind is that we will need to synthesize a continuum of stimuli if we are to carry out similar experiments. That may be done in Praat. In Praat one can create both a source signal (or extract one from an existing sound) and a filter (or extract one from an existing sound). Both the source signal and the filter may be manipulated (search for "source-filter synthesis" in Praat's Help manual). The source is generated by creating and/or modifying a PitchTier object and then deriving a PointProcess object. Finally, one creates a Sound object from the modified PointProcess object. The filter is generated with Praat's FormantGrid object. This may be modified manually or with a script. The last step is done by filtering the source Sound object through the FormantGrid object. A second possibility is to use a plug-in package that has been developed for Praat. Bartlomiej Plichta has written Akustyk, a powerful, free and multi-platform Praat add-on (bartus.org/akustyk/). The advantage of Akustyk is that its author has gone to great lengths to make the process of synthesizing speech (as well as other tasks) relatively simple for the apprentice. For instance, he has posted a number of video tutorials that teach how to use Akustyk, including how to synthesize continua. Synthesized continua may be created with just a few clicks in Akustyk without needing to learn about sources and filters.

How may listener responses be collected? There are many ways to do that. In addition to the psychology software packages mentioned above, one might decide to use Praat. Praat's ExperimentMFC object is relatively intuitive to learn. With this object one can develop simple identification and discrimination experiments. Listeners may respond to auditory stimuli by using the mouse and clicking on boxes drawn on the screen. A second possibility is to configure the keyboard so that listeners may introduce their responses by pressing different keys, e.g. left-shift for 'different' and right-shift for 'same.'

Pallier, Colomé & Sebastián-Gallés (2001) is a follow-up of the experiments discussed above. The authors investigated the effects of dominant-specific categorical perception in lexical access; that is, they studied whether the fact that Catalan-dominant bilinguals show categorical perception for Catalan-specific vowel contrasts while Spanish-dominant bilinguals do not would have an effect on their patterns of spoken word recognition. Bilingual listeners were provided with the following types of materials: (1) a set of Catalan minimal pairs that differed in a Catalan-specific vowel contrast difficult to perceive by Spanish-dominant listeners, (2) a set of pseudo-word minimal pairs differing in the same type of vowel contrast and (3) a set of word and pseudo-word fillers. Listeners were to hear stimuli one at a time and were to respond whether they had heard a real Catalan word or a non-word. Response accuracy as well as time to reaction (i.e. latency between auditory stimuli and response) were collected. The design was based on the concept of *medium-term auditory repetition priming*; that is, the concept that it is faster to recognize a word when the word has been heard recently (i.e. primed). In particular, the authors reasoned that the words involved in a Catalan-specific minimal pair will not prime each other for Catalan-dominant listeners (since these are not “repetitions”) while they might for Spanish-dominant listeners (who might consider them “repetitions” since they have difficulties processing the acoustic differences of the members of the pair). The authors created lists in which a member of a minimal pair would be followed, 8 to 20 items further down the list, by the other member of the pair or by itself.

This experiment illustrates a specific use of technology in perception research, i.e. the use of software that is capable of recording reaction times. This, apparently, is more difficult than what it might seem due to the fact, among other things, that computer-keyboard timing relations are complex. In other words, the keyboard takes a few ms to send the signal to the computer and this delay seems to vary almost randomly. The scientific community has come up with software packages that accurately record time (such as those mentioned above: SuperLab, E-Prime, PsyToolkit, etc.) and hardware pieces that communicate with the computer with a minimum, non-random delay. These are called button- or response-boxes or response-pads. By means of a conclusion to this section, consider what Pallier, Colomé & Sebastián-Gallés (2001) found: Catalan-dominant listeners did not display repetition priming for minimal pairs while Spanish-dominant listeners did. In other words, Spanish-dominant listeners implicitly treated the two members of the minimal pairs as homophones. Interesting, isn’t it? As Johnson (2003, p.60) puts it: “There is no end to how clever people can be when it comes to devising listening experiments.” Access to the right tools, as much as cleverness, is key to making all this possible.

4. Setting up a basic phonetics laboratory

This section provides some information about how to set up a basic phonetics laboratory for acoustic and auditory-perceptual research. I offer some remarks about hardware; that is, the equipment one would need to buy to mount the laboratory. The idea was inspired by Marinis' (2003) article in Second Language Research, in which the author gives some advice on how to set up a psycholinguistics laboratory.

It is customary to collect data both in the laboratory and in the field. The equipment needs are, for the most part, different. In the laboratory the emphasis is put on the quality of the equipment while in the field the emphasis is put on its portability. However, the only necessary difference between these two settings is that, space and budget permitting, one would want to purchase a sound-treated booth (WhisperRoom, VocalBooth) or anechoic chamber (IAC) for the laboratory. This improves the quality of the data significantly and should be considered by anyone with the sufficient budget and space. Regarding hardware, one could use the same equipment in the laboratory and in the field as long as he or she would take good care to select portable, yet professional-quality, equipment. The remainder of the section focuses on data collection in the field.

Bartlomiej Plichta maintains a site in which he provides some advice on field recording techniques and posts reviews of equipment, including microphones, pre-amplifiers, voice recorders and audio interfaces (bartus.org/akustyk/signal.php). I sincerely recommend reading his advice and reviews. He insists mostly on developing a good recording technique rather than on purchasing the most expensive equipment. I here simply add some remarks.

In order to collect production (acoustic) data in the field one would need the following equipment: (1) a microphone, (2) a microphone pre-amplifier, and (3) a digital voice recorder. An alternative to the digital voice recorder would be to obtain an audio interface (Firewire or USB) that would connect to a laptop and carry out the analog-to-digital conversion. The laptop could be running Praat or Audacity to obtain the recording. Let us start with the microphone. This is an essential piece of equipment. There are fundamentally two types of microphones in the market: dynamic and condenser. Condenser microphones need phantom power; that is, they need to be connected to a power source. Most pre-amplifiers, recorders and audio interfaces are able to provide power to condenser microphones but many portable recorders and interfaces have low quality phantom power. This is a constant source of noise that affects the quality of the signal significantly. One lesson that I have learned over the years is that professional microphone pre-amplifiers can solve this problem and are thus exceptional pieces of equipment. One could connect the microphone directly to the voice recorder or the audio interface. However, there is a tremendous difference between signals obtained with or without a pre-amplifier. Dynamic microphones do not need to be connected to an energy source, so one would turn off the phantom power switch of the pre-amplifier, recorder or interface.

A number of researchers recommend condenser microphones because they are less prone to recording surrounding noise and they are more sensitive. Of course, that will depend on the quality of the microphone itself and also on the quality of the rest of the equipment.

In the field it is better to use head-mounted microphones. These microphones are head-worn by the speaker so that he or she may freely move without altering the quality of the signal. The microphone bug remains at a constant distance from the speaker's lips. Although head-mounted, they are minimally invasive and most speakers do not appear to be disturbed by them.

My personal opinion is that investing in high-quality recording equipment (a good microphone, a good pre-amplifier, a digital voice recorder or a good USB/Firewire audio interface) is much more important than investing in commercial software. The main reason for this is that there are free "versions" of most commercial software packages while there are no free or cheap equivalents to recording equipment. The quality of the tools significantly affects the quality of the job, as any professional of any field would confirm. At any rate, I wish to insist here that investing in good recording equipment is not an option at this juncture, as the quality of signals obtained directly through personal computers, small MP3 recorders or cheap table microphones is not acceptable anymore and professional-quality equipment is mostly affordable.

What does one need for field perception research? The following instruments: (1) good head-phones, (2) a response-pad, (3) dedicated software, and (4) a laptop computer. If reaction times are not critical for one's research, one might save the money of the dedicated software and the response-pad, but not that of the laptop computer or the professional studio head-phones. In this case, one might decide to use Praat to design and run experiments. On the other hand, if reaction times are important, it is necessary to invest in the appropriate software (SuperLab, E-Prime, Presentation, Paradigm, etc.) and a response-pad (Cedrus, IoLab Systems). A third possibility is to run the experiments with free experimental design software, such as DMDX (Windows), PsyScope X (Mac OS X) or PsyToolkit (Linux). However, there is no alternative to purchasing a response-box. This is a necessary piece of equipment. One might also want to purchase a USB or Firewire audio interface to carry out the digital-to-analog signal conversion.

The present article has discussed some aspects of the use of technology in modern experimental phonetics. Instead of surveying the different instruments that are being used, I have decided to discuss the capabilities of some widely-available tools, some of which are free. In particular, I discussed some uses of Praat (acoustic and perceptual research) and R (statistics) while providing examples from the literature. Of course, many other instruments are available. However, in my experience, it is at least as wise to spend some time learning the full potential of the tools one already has or one may acquire easily than constantly peeking into different techniques without doing more than scratching the surface. But perhaps this is my bias. Finally, I commented on the type of equipment one would need to purchase in order to set up a basic acoustic-auditory laboratory or, more specifically, in order to use the right tools to carry out phonetic fieldwork. In this regard, my main point has been that, in order to produce serious research, one needs to use serious equipment. Serious equipment is now relatively affordable and widely available.

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